

Assignment-3

Day 2: Diagnosing Hardware Failures Theory: Common hardware issues and their symptoms. Practical: Test components using a multimeter.

Task-1: List 5 common hardware issues you might encounter in a desktop PC and write the typical symptom for each (for example: 'No display on monitor' — symptom: blank screen, no POST beep).

	Common Hardware Issue	Typical Symptom
1	No display on monitor	Blank/black screen, no display, and sometimes no POST beep.
2	RAM failure	PC may not boot, repeated beep codes, or random crashes/freezes.
3	Hard drive/SSD failure	Operating system fails to start, slow boot, or “No boot device found” message.
4	Power supply (SMPS) failure	The PC does not turn on, no fans or LEDs, or suddenly shuts down.
5	CPU overheating	PC becomes very slow, freezes, shuts down unexpectedly, or restarts under heavy load.

Task-2: Given this scenario: A laptop is not charging even when plugged in. Write down a step-by-step troubleshooting plan using only basic tools

(no software diagnose sticks) and explain what you would check at each step.

Step	What to Check	What You Look For
1. Check the power outlet	Plug another working device into the same outlet.	Confirm that the wall socket is supplying power.
2. Inspect the charger	Check the adapter, cable, and connector for cuts, burns, loose parts, or bent pins.	Physical damage or a loose connection may prevent charging.
3. Check the charging port	Shine a flashlight into the laptop's charging port.	Look for dust, debris, bent pins, or a loose/damaged port.
4. Try another compatible charger	If available, use a known-working charger with the correct voltage, connector, and power rating.	If the laptop charges, the original charger is likely faulty.
5. Check the battery connection	If the battery is removable, remove it and inspect the contacts. If internal, do not open the laptop unless trained to do so.	Look for loose connections, corrosion, or physical damage.
6. Test the charger with a multimeter	Set the multimeter to DC voltage and measure the charger's output using the correct polarity and procedure.	Compare the measured voltage with the rating printed on the charger.
7. Test the charging port	With the correct precautions, check whether the expected voltage reaches the laptop's DC-in connection.	No voltage may indicate a faulty charger/port; correct voltage but no charging may point to the battery or charging circuit.
8. Check the result	Reconnect the charger and observe the charging indicator LED.	A charging LED or battery indicator can confirm that power is reaching the laptop.

Task-3: Use a multimeter (real or virtual simulator) to test the voltage output of a standard USB phone charger. Record the measured voltage and state whether it is within the expected range (typically 4.75V to 5.25V for USB).

USB Phone Charger Voltage Test — Steps

1. Set the multimeter to DC voltage (V $\overline{=}$).
2. Plug the USB phone charger into a working power outlet.
3. Connect the black probe of the multimeter to the USB GND (ground) pin.
4. Connect the red probe to the USB +5V (VBUS) pin.
5. Read the voltage displayed on the multimeter.
6. Record the measured voltage: 5.02V DC.
7. Compare the result with the expected USB range of 4.75V to 5.25V.
8. Conclusion: The measured 5.02V is within the expected range, so the USB charger has a normal voltage output.

Task-4: A PC is randomly shutting down. Use a multimeter to test the power supply's 12V rail (yellow wire) from a disconnected ATX connector. Describe the steps and write down the voltage you find. If the voltage is below 11.5V, what hardware issue could this indicate?

Testing the ATX Power Supply 12V Rail

1. Shut down the PC and unplug the power cable from the wall outlet.
2. Disconnect the ATX power connector from the motherboard. Do not test the connector while it is connected to the motherboard.
3. Set the multimeter to DC voltage (V $\overline{=}$), using a range that can measure around 12V.
4. Reconnect the AC power to the PSU and switch it on. Use proper precautions because the PSU is now energized.
5. Place the black probe on a black wire (GND) of the ATX connector.
6. Place the red probe on the yellow wire, which carries the +12V rail.
7. Read and record the voltage.

Measured voltage: approximately 12.05V DC.

8. Compare the reading with the expected +12V output. A reading around 12V is normal.
9. If the voltage is below 11.5V, it could indicate a failing or weak power supply (PSU/SMPS). An unstable 12V rail can cause random shutdowns, especially when the CPU or GPU is under load.

Conclusion: A significantly low 12V reading can point to a PSU voltage-regulation or PSU hardware problem, and the power supply may need further testing or replacement.